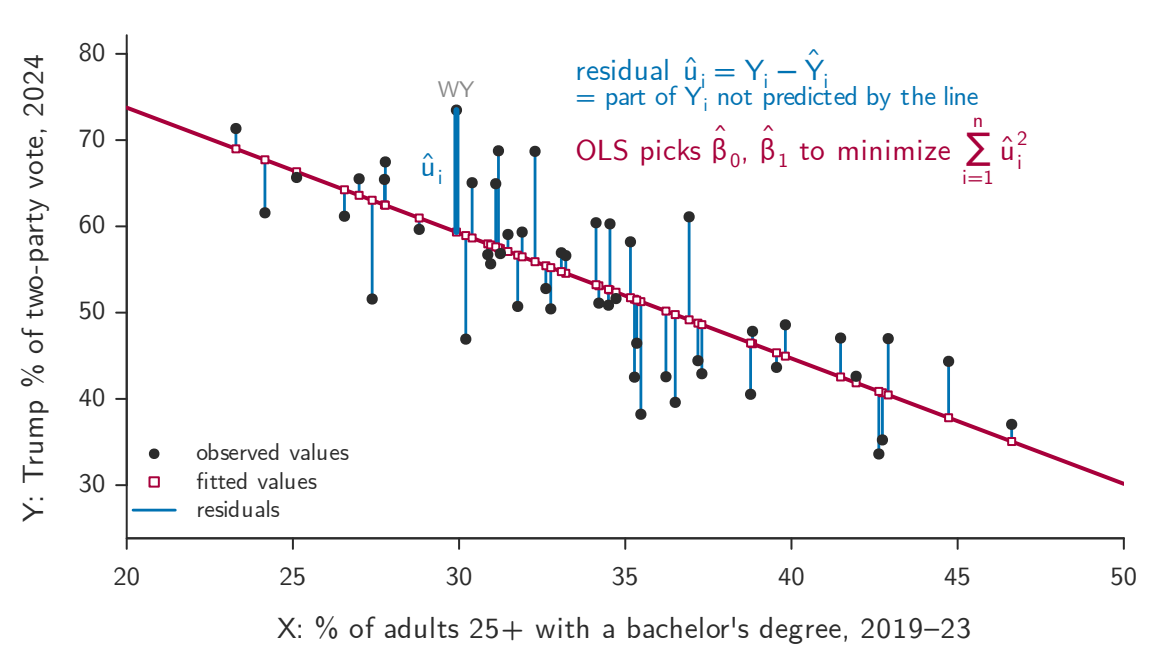
residual $\hat{u}_i = Y_i - \hat{Y}_i$
= part of Y_i not predicted by the line
OLS picks $\hat{\beta}_0$, $\hat{\beta}_1$ to minimize $\sum_{i=1}^{n} \hat{u}_i^2$
WY
$\hat{u}_i$
observed values
fitted values
residuals
Y: Trump % of two-party vote, 2024
X: % of adults 25+ with a bachelor's degree, 2019–23
80
70
60
50
40
30
20
25
30
35
40
45
50